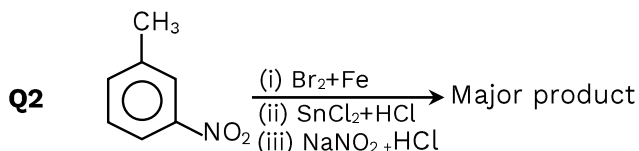


CHEMISTRY

- Q1** Which of the following complex shows facial-meridional isomerism
 (1) $[\text{Cr}(\text{en})_3]^{3+}$ (2) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ (3) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ (4) $[\text{Cr}(\text{en})_2\text{Cl}_2]^+$

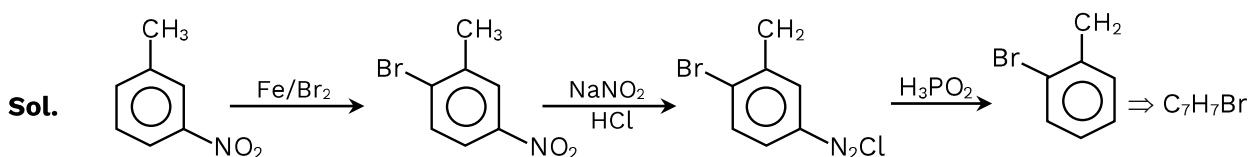
Ans. (4)

Sol. Its $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
 So, X how facial & meridional we
 Need $[\text{MA}_3\text{B}_3]$ type complex



Find the molecular weight of the major product

Ans. (159)

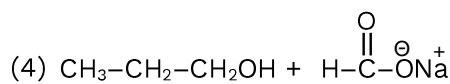
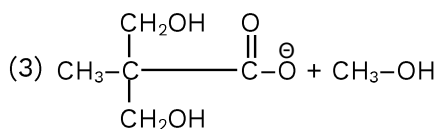
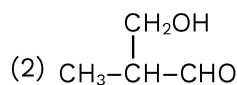
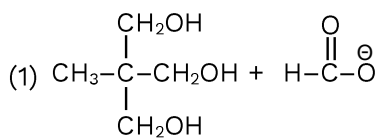
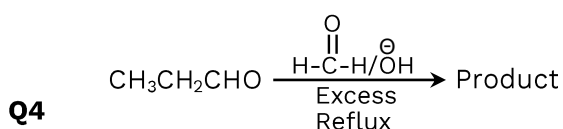


Molecular weight - $\text{C}_7\text{H}_7\text{Br}$
 $= 7 \times 12 + 7 \times 1 + 80$
 $= 72 + 7 + 80$
 $= 159$

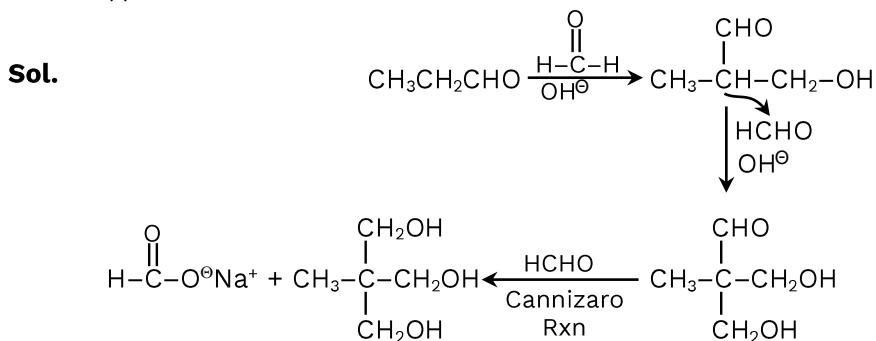
- Q3** Which of the following does not belong to the same period?
 (1) Os (2) Ir (3) Pd (4) Pt

Ans. (3)

Sol. Factual



Ans. (1)



Q5 Which of the following will react with Hinsberg reagent?

(A) Aniline

(B) N,N-Dimethyl aniline

(C) $C_6H_5NHC_6H_5$

(D) Methyl amine

(1) A and C only

(2) A, C and D only

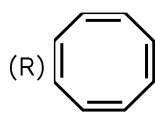
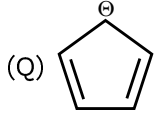
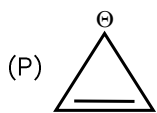
(3) A only

(4) A and B only

Ans. (1)

Sol. Primary and secondary amine can react with Hinsberg reagent due to acidic H of NH.

Q6 Identify the stability order



(1) $Q > R > P$

(2) $R > P > Q$

(3) $Q > P > R$

(4) $P > Q > R$

Ans. (1)

Sol. In option (1) Carbanion is stable due to aromaticity of ring.

Q7 Given below are two statements:

Statement-I: Fructose can give Tollen's test even though it does not have aldehyde group

Statement-II: When reacted with base fructose can undergo rearrangement to produce aldehyde group.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

Ans. (3)

Sol. Conceptual

Q8 Which of the following pairs of ions (aq. solution) show same colour?

(1) V^{2+} , Cr^{3+} , Mn^{3+}

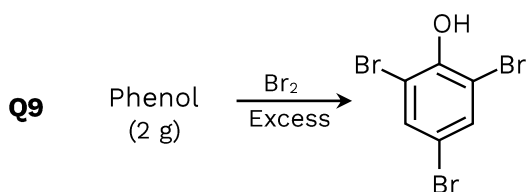
(2) Zn^{2+} , Cr^{3+} , Mn^{2+}

(3) Sc^{3+} , Ti^{3+} , V^{3+}

(4) Ti^{4+} , Mn^{2+} , Cr^{2+}

Ans. (1)

Sol. NCERT based.



How many gram Bromine is required ?

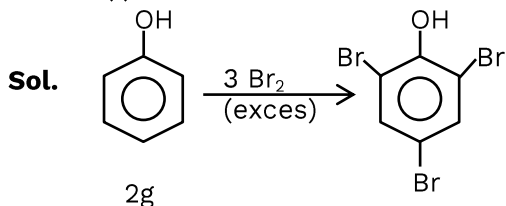
(1) 10.22 g

(2) 6 g

(3) 4 g

(4) 12.22 g

Ans. (1)



1 mole phenol need 3 mole Br_2

so 2 gm ie $\frac{2}{94}$ mole need $\frac{2}{94} \times 3$

$$\text{Mans of Br}_2 = \frac{2}{94} \times 3 \times 160 \approx 10.22$$

Nearest integer = (10)

Q10 Match the List:

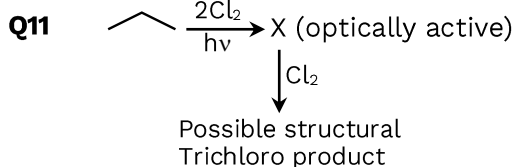
List-I

- (P) Octet complete
 (Q) Incomplete octet with odd e^-
 (R) Incomplete octet
 (S) Expanded octet
 (1) P→3; Q→4; R→1; S→2
 (3) P→1; Q→3; R→4; S→2

List-II

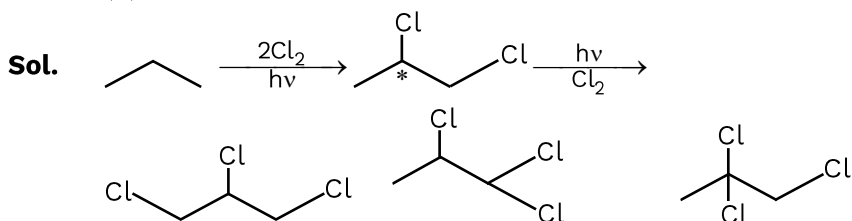
- (1) BF_3 , AlCl_3
 (2) PF_5 , H_2SO_4
 (3) CO_2 , CCl_4
 (4) NO , NO_2
 (2) P→4; Q→3; R→1; S→2
 (4) P→1; Q→3; R→2; S→4

Ans. (1)



- (1) 2 (2) 3 (3) 6 (4) 5

Ans. (2)



Q12 Magnetic moment of a complex is 3.87 B.M. has Co^{2+} metal ion than number of electrons in t_{2g} and e_g orbital in octahedral complex

- (1) t_{2g}^5, e_g^2 (2) t_{2g}^6, e_g^1 (3) t_{2g}^4, e_g^3 (4) t_{2g}^3, e_g^4

Ans. (1)

Sol. $\mu = 3.87$ B.M.

Number of unpaired $e^- = 3$



Q13 Which of the following statement is incorrect

- (1) PH_3 has lower proton affinity than NH_3
 (2) SO_2 act as oxidising agent but not as reducing agent
 (3) NO_2 dimerises easily to N_2O_4
 (4) PF_5 exist but NF_5 does not exist.

Ans. (2)

Q14 Match the following

Name

Product

(P) Swartz reaction

(A) Ethyl Flouride

(Q) Wurtz-Fittig reaction

(B) Ethyl Iodide

(R) Finkelstein reaction

(C) Ethyl Benzene

(S) Sandmayer reaction

(D) Cyano Benzene

(1) P→A, Q→C, R→B, S→D

(2) P→B, Q→C, R→A, S→D

(3) P→D, Q→B, R→A, S→C

(4) P→A, Q→D, R→C, S→B

Ans. (1)

Q15 Consider the given values:

$$\Delta H = 55 \text{ kJ mol}^{-1}$$

$$\Delta S = 175 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$T = 25^\circ\text{C}$$

Calculat the value of Gibbs free energy charge (ΔG) in J mol^{-1}

Ans. (2.85×10^{-3})

Sol. $\Delta G = \Delta H - T\Delta S$

$$= 55 \times 10^{-3} - (25 + 273) (175)$$

$$= 55 \times 10^{-3} - 298 \times 175$$

$$= 55 \times 10^{-3} - 52.150 \times 10^{-3}$$

$$= 2.85 \times 10^{-3}$$

Q16 In estimation of sulphur by Carius method, 160 gm of organic compound gives 466 gm of Barium sulphate. % of sulphur in the organic compound is:

Ans. (40)

Sol. (40%) 160g of compound $\rightarrow$ 466 g BaSO_4



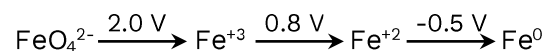
$$\begin{array}{ccc} \boxed{\text{OC}} & \xrightarrow{\quad} & \text{BaSO}_4 \\ 160 \text{ gm} & & 466 \text{ gm} \end{array}$$

$$\% \text{ S} = \frac{2 \times 32}{160} \times 100 \quad \frac{466}{233} = 2 \text{ mole} = n_{\text{sulphur}}$$

$$= 40 \%$$

$$m_s = 2 \times 32 \text{ gm}$$

Q17 consider the following



$$E^0_{\text{FeO}_4^{2-}/\text{Fe}^{2+}}$$

Ans. (2)

Sol. $E^\circ = \frac{n_1 E_1 + n_2 E_2}{n_1 + n_2}$

$$= \frac{3 \times 2 + 1 \times 0.8}{4} \Rightarrow 1.7 \text{ V}$$

Nearest Integer = 2

Q18 Given below are two statements:

Statement-I: During Lassaigne's test, covalent compound is converted to ionic compound.

Statement-II: $\text{Na}_4[\text{Fe}(\text{CN})_6]$ gives prussian blue colour on reaction with $\text{Fe}_2(\text{SO}_4)_3$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Both Statement I and Statement II are correct
- (4) Statement I is incorrect but Statement II is correct

Ans. (1)

Q19 If 10^{21} molecules are removed from x mg of $\text{CO}_2(\text{g})$, then 2.8×10^{-3} moles are left.

Then calculate the value of x ?

Ans. (1179)

Sol.
$$\frac{x \times 10^{-3}}{44} \times 6 \times 10^{23} - 10^{21} = 2.8 \times 10^{-3} \times 6 \times 10^{23}$$
$$\frac{x \times 10^{20}}{44} - 10^{21} = 2.8 \times 6 \times 10^{20}$$
$$\frac{x}{44} - 10 = 2.8 \times 6$$
$$x = (2.8 \times 6 + 10)44$$
$$= 1179.2 \text{ mg} \approx 1179$$

Q20 Calculate the value of X in compound $\text{CrCl}_3 \cdot X \text{NH}_3$ if elevation in boiling point of solution is 0.558°C and molality of solution is 0.1. Given $K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C Kg-mol}^{-1}$

- (1) 6 (2) 7 (3) 5 (4) 2

Ans. (3)

Sol. $\Delta T_f = 0.558^\circ\text{C}$

$$K_f = 1.86$$

$$m = 0.1$$

$$\Delta T_b = i \times K_b \times m$$

$$0.558 = i \times 1.86 \times 0.1$$

$$i = \frac{0.558}{0.186} = 3$$

